DERIVED-CATEGORY EQUIVALENCE FOR CARTESIAN MODULES OVER A HYPERCOVERING: LA^* WITH QUASI-INVERSE RA_*

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the parenthetical directing the reader to the modules-versus-abelian-sheaves cohomology comparison (source lines 5520-5521) is a supplementary cross-reference, not an omission; the covering argument for the essential image is written out in full.

Editorial note. The label lemma-hypercovering-equivalence-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Cartesian modules form a weak Serre subcategory. Same chapter, \label{lemma-Serre-subcat-cartesian-modules}. Lines 5457-5460: the base weak Serre fact, its hypotheses holding by the discussion in Remark remark-augmentation-ringed.
- The bounded modules equivalence for hypercoverings. Same chapter, \label{lemma-hypercovering-equivalence-bounded-modules}. Line 5468: La_*a^*F = F, the unit half carried over from the bounded version.
- Module-level descent: cartesian modules are pulled back. Same chapter, \label{lemma-hypercovering-descent-modules}. Line 5474: every cartesian O-module G is a^*F for some F.
- The weak Serre conditions on the A_U. Same chapter, conditions (\ref{item-restriction}), (\ref{item-local}), (\ref{item-bounded-dimension}). Lines 5490-5494 and 5510-5519: restriction and localness close the essential-image argument; bounded-dimension feeds assumptions (4) and (5) of the criterion.
- Cohomology on Sites, the unbounded equivalence criterion. Cohomology on Sites chapter, \label{sites-cohomology-lemma-equivalence-unbounded-one}. Line 5497: the five-assumption criterion that upgrades the abelian-category equivalence to the derived equivalence.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Cohomology of modules agrees with cohomology of abelian sheaves. Cohomology on Sites chapter, \label{sites-cohomology-lemma-cohomology-modules-abelian-agree}. Lines 5520-5521: supplementary cross-reference for the reader worried about the module vs abelian sheaf distinction.

Source reference. Original reference 0DC4: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5317–5317. Prerequisite pointer invoked at proof line(s) 5437; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DC3: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5310–5310. Prerequisite pointer invoked at proof line(s) 5437; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DC2: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5305–5305. Prerequisite pointer invoked at proof line(s) 5436; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7J: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2669–2669. Prerequisite pointer invoked at proof line(s) 5457; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9Z: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4692–4692. Prerequisite pointer invoked at proof line(s) 5476; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DA2: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4754–4754. Prerequisite pointer invoked at proof line(s) 5470; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DC0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4016–4016. Prerequisite pointer invoked at proof line(s) 5513; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9W: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 4073–4073. Prerequisite pointer invoked at proof line(s) 5441; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0D8A: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3902–3902. Prerequisite pointer invoked at proof line(s) 5507; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03FD: sources/stacks-chow/tags.txt, source lines 4488–4488. Prerequisite pointer invoked at proof line(s) 5520; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D7V: sources/stacks-chow/tags.txt, source lines 16570–16570. Prerequisite pointer invoked at proof line(s) 5498; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O}_\mathcal{C})$ be a ringed site. Assume given weak Serre subcategories $\mathcal{A}_U \subset Mod(\mathcal{O}_U)$ satisfying conditions (0DC2), (0DC3), and (0DC4) above. Assume $\mathcal{C}$ has equalizers and fibre products and let K be a hypercovering. Let $((\mathcal{C}/K)_{total}, \mathcal{O})$ be as in Remark 0D9W. Let $\mathcal{A}_{total} \subset Mod(\mathcal{O})$ denote the weak Serre subcategory of cartesian $\mathcal{O}$-modules $\mathcal{F}$ whose restriction $\mathcal{F}_n$ is in $\mathcal{A}_{K_n}$ for all n (as defined above). Then the functor La^* defines an equivalence*

$$D_\mathcal{A}(\mathcal{O}_\mathcal{C}) \longrightarrow D_{\mathcal{A}_{total}}(\mathcal{O})$$

with quasi-inverse Ra_*.

Proof. The cartesian $\mathcal{O}$-modules form a weak Serre subcategory by Lemma 0D7J (the required hypotheses hold by the discussion in Remark 0D9W). Since the restriction functor $g_n^* : Mod(\mathcal{O}) \to Mod(\mathcal{O}_n)$ are exact, it follows that $\mathcal{A}_{total}$ is a weak Serre subcategory.

Let us show that $a^* : \mathcal{A} \to \mathcal{A}_{total}$ is an equivalence of categories with inverse given by La_*. We already know that $La_*a^*\mathcal{F} = \mathcal{F}$ by the bounded version (Lemma 0DA2). It is clear that $a^*\mathcal{F}$ is in $\mathcal{A}_{total}$ for $\mathcal{F}$ in $\mathcal{A}$. Conversely, assume that $\mathcal{G} \in \mathcal{A}_{total}$. Because $\mathcal{G}$ is cartesian we see that $\mathcal{G} = a^*\mathcal{F}$ for some $\mathcal{O}_\mathcal{C}$-module $\mathcal{F}$ by Lemma 0D9Z. We want to show that $\mathcal{F}$ is in $\mathcal{A}$. Take $U \in \mathrm{Ob}(\mathcal{C})$. We have to show that the restriction of $\mathcal{F}$ to $\mathcal{C}/U$ is in $\mathcal{A}_U$. As usual, write $K_0 = \{U_{0,i}\}_{i \in I_0}$. Since K is a hypercovering, the map $\coprod_{i \in I_0} h_{U_{0,i}} \to *$ becomes surjective after sheafification. This implies there is a covering $\{U_j \to U\}_{j \in J}$ and a map $\tau : J \to I_0$ and for each $j \in J$ a morphism $\varphi_j : U_j \to U_{0,\tau(j)}$. Since $\mathcal{G}_0 = a_0^*\mathcal{F}$ we find that the restriction of $\mathcal{F}$ to $\mathcal{C}/U_j$ is equal to the restriction of the $\tau(j)$th component of $\mathcal{G}_0$ to $\mathcal{C}/U_j$ via the morphism $\varphi_j : U_j \to U_{0,\tau(i)}$. Hence by (0DC2) we find that $\mathcal{F}|_{\mathcal{C}/U_j}$ is in $\mathcal{A}_{U_j}$ and in turn by (0DC3) we find that $\mathcal{F}|_{\mathcal{C}/U}$ is in $\mathcal{A}_U$.

In particular the statement of the lemma makes sense. The lemma now follows from Cohomology on Sites, Lemma 0D7V. Assumption (1) is clear (see Remark 0D9W). Assumptions (2) and (3) we proved in the preceding paragraph. Assumption (4) is immediate from (0DC4). For assumption (5) let $\mathcal{B}_{total}$ be the set of objects $U/U_{n,i}$ of the site $(\mathcal{C}/K)_{total}$ such that $U \in \mathcal{B}$ where $\mathcal{B}$ is as in (0DC4). Here we use the description of the site $(\mathcal{C}/K)_{total}$ given in Section 0D8A. Moreover, we set $\mathrm{Cov}_{U/U_{n,i}}$ equal to Cov_U and $d_{U/U_{n,i}}$ equal d_U where Cov_U and d_U are given to us by (0DC4). Then we claim that condition (5) holds with these choices.

This follows immediately from Lemma 0DC0 and the fact that $\mathcal{F} \in \mathcal{A}_{total}$ implies $\mathcal{F}_n \in \mathcal{A}_{K_n}$ and hence $\mathcal{F}_{n,i} \in \mathcal{A}_{U_{n,i}}$. (The reader who worries about the difference between cohomology of abelian sheaves versus cohomology of sheaves of modules may consult Cohomology on Sites, Lemma 03FD.) $\square$